

No.	Solution
4(a)	By Newton's law of gravitation, gravitational force between two point masses M and m, $F = \frac{GMm}{r^2}, \text{ where } G \text{ is the gravitational constant.}$ Gravitational field strength is the gravitational force per unit mass acting at a point in the gravitational field. $g = \frac{F}{m}$ $= \frac{\frac{GMm}{r^2}}{m}$ $= \frac{GM}{r^2} \text{ (shown)}$ <i>Note: A common mistake is not giving Newton's law of gravitation or the definition of g, as required by the question.</i>
4(b)	Gravitational force on the satellite by Earth provides for its centripetal force in orbit. $\frac{GMm}{r^2} = mr\omega^2$ $= mr \left(\frac{2\pi}{T} \right)^2$ $r^3 = \frac{GM}{4\pi^2} T^2$ $= \frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{4\pi^2} (110 \times 60)^2$ $r = 7.615 \times 10^6 \text{ m}$ Gravitational field strength, $g = \frac{GM}{r^2}$ $= \frac{(6.67 \times 10^{-11})(6.0 \times 10^{24})}{(7.615 \times 10^6)^2}$ $= 6.901$ $\approx 6.90 \text{ N kg}^{-1}$
4(c)	The gravitational force by the Earth on an object of mass m is given by the value mg, where g is the gravitational field strength. By Newton's second law, this force is the weight of the object that gives rise to its resultant acceleration of free fall a, resulting in $mg = ma$. When near the surface of the Earth, the gravitational field strength $g = \frac{GM}{r^2}$, where G is gravitational constant, M the mass of Earth and r

No.	Solution
	the distance between the centres of masses of the object and the Earth, the value r remains approximately equal to the radius of the Earth, making the gravitational field strength equal to the acceleration of free fall. <i>Note: Explanation needs to make the link between the definition of g and the meaning of “acceleration of free fall”.</i>
5(a)	Using $v = f\lambda$